AERM Exam Outputs and Results

2026-03-23

Part 1

Use Julia code

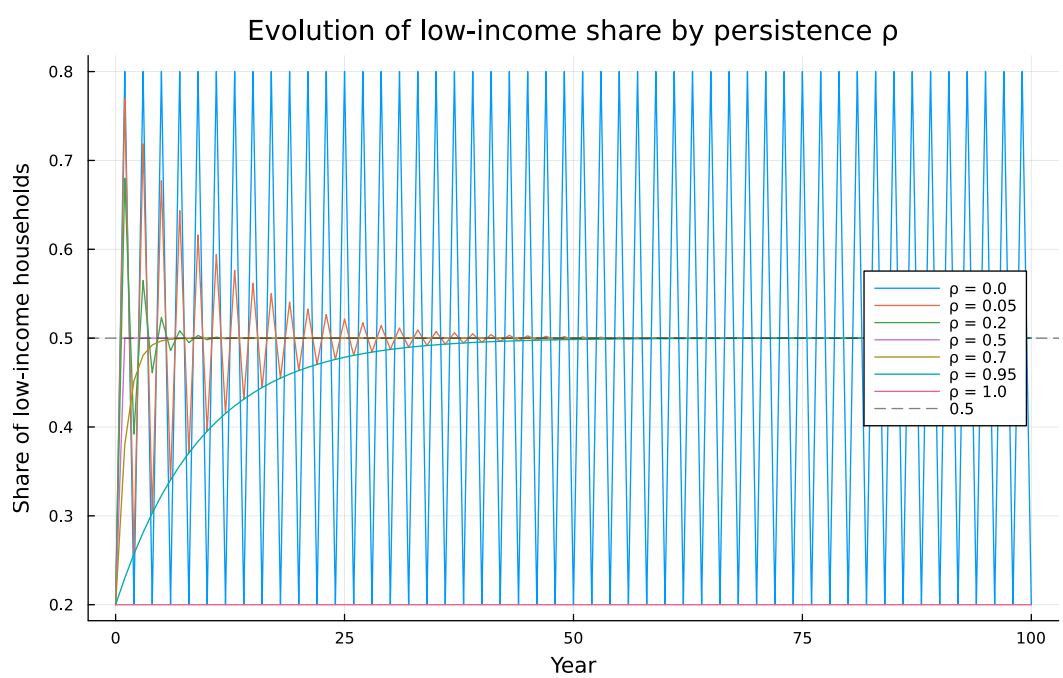
Exercise 1: Solve Standard Consumption-Saving Problem for Different Values of ρ

Q1: Stationary Income Distribution

Income follows a two-state Markov chain with symmetric persistence $\rho_L = \rho_H = \rho$ and transition matrix:

$$P = \begin{pmatrix} \rho & 1-\rho \\ 1-\rho & \rho \end{pmatrix}$$

Starting from $P(y = y_L) = 0.2$, the figure below shows the share of low-income households over a 100-year horizon for each value of ρ.



Explanation:

For all $\rho \in (0, 1)$, the share of low-income households converges to 0.5, which is the unique stationary distribution of this symmetric Markov chain, but higher persistence slows the rate of convergence.

The two edge cases behave differently: $\rho = 0$ causes full state-switching each period (so the share oscillates before settling), while $\rho = 1$ makes both states absorbing, so the initial distribution $P(y_L) = 0.2$ is permanent.

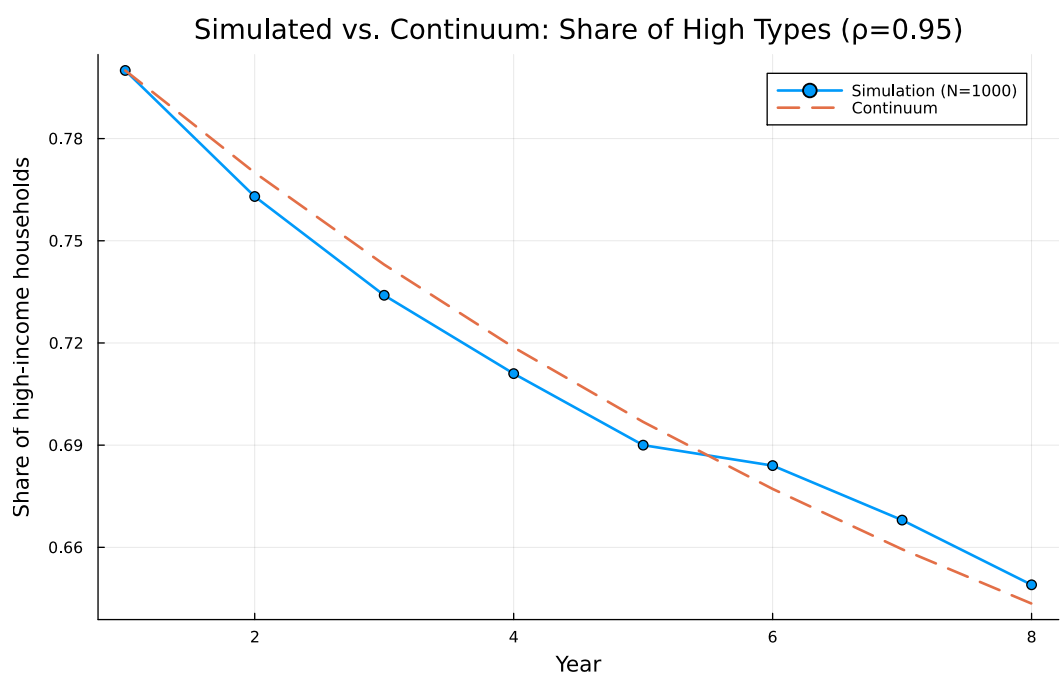
Q2: Volatility vs. Risk

Explanation:

In the stationary distribution, the variance of $\Delta \log y$ equals $(1 - \rho) \cdot (\log 2)^2$, which is maximized at $\rho = 0$ where every household switches state each period.

However, $\rho = 0$ does not generate the most risk for households: with i.i.d. income, bad shocks are transitory and easily self-insured through savings, whereas high persistence (e.g. $\rho = 0.95$) means adverse shocks last many periods, making them far harder to smooth.

Q3: Simulation vs. Continuum



The simulated share of high-income types closely tracks the continuum prediction but does NOT match it exactly, because a finite sample of 1000 households is subject to sampling variation whereas the continuum result holds by the law of large numbers over a unit mass of agents.

Q4: OLS vs. Fixed Effects

Using the simulated panel data, I regress current income on lagged income via OLS and fixed effects (implemented by demeaning), using only linear algebra as instructed.

Model	$\hat{\beta}$
OLS	0.8983
FE	0.5690
True ρ	0.95

OLS is closer to the true $\rho = 0.95$ because the data-generating process contains no individual fixed effects — all households face the same Markov chain — so OLS is consistent. The FE estimator suffers from Nickell bias: with only $T = 8$ periods, demeaning mechanically creates a negative correlation between the demeaned lagged variable and the demeaned error, biasing $\hat{\beta}_{FE}$ substantially downward. This bias is systematic and would shrink only as $T \to \infty$.